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Information Systems Analysis and Design

ISQS-4349-001

Project 1: System Request

Showtime Sound & Lighting (SSL) Inc

Executive Summary

Showtime Sound and Lighting (SSL) is a growing business that is owned by Tom Smith. SSL provides lighting and sound services for local entertainment companies and other civic groups that need additional support for larger gatherings. These services include DJ service, equipment rental, and sale of audio, video, and lighting equipment.

SSL began to experience a steady decrease in productivity. This can be attributed to the current information system that is recorded manually. In addition to being recorded manually, the current system has issues with data duplication, poor time management, and inability to effectively meet the demands of a growing list of customers. Due to the current system, SSL is affecting the service quality, causing loss of repeat customers, bad customer service, affecting the sustained growth of the company.

To address these issues, SSL would like to implement a new, automated information system to make data entry more efficient, improve accuracy and efficiency of customer and equipment records, and improve inventory accuracy and records. This new system would result in more accurate information, more time for employees to perform other tasks, more efficient retrieval of data, and better customer service provided, increasing company sustainability.

This document includes a systems request, describing the project sponsor, business needs and requirements, potential value of the project, and issues or constraints raised by enacting the project. This request will include Functional and non-functional requirements, use case diagrams, activity diagrams, use case descriptions, ER diagrams, sequence diagram and a user interface.

System request for Showtime Sound & Lighting (SSL) Inc

System Request - SSL Integration of Information System Development

Project Sponsor - Tom Smith, CEO, SSL Inc.

Business Need: SSL Inc. has experienced a tremendous growth in business. It has seen a decrease in productivity in business due to inefficient information system and manual and outdated information systems. The system lacks integration and has led to several mistakes such as duplication of data, difficult in accessing certain important information on time, and the decline of productivity and customer service quality. Mr. Smith has identified a need for a more up to date system to better integrate business data and an automated information system that will support current and future growth of the business. If this isn't done Mr. Smith plans on losing about 5-10% of his customers in the next two years.

Business Requirements: The new system must maintain digital information about new and old customers to provide information when needed for decision making by management. It needs to be able to track equipment inventory such as availability, and maintenance. It has to schedule equipment and personnel for customer event. The system to report analytics for decision making. It also needs to track customers history and past jobs. The system needs to generate and manage proposals, orders and invoices electronically. It needs to also manage contract workers assignments and evaluations. System must handle growth to accommodate up to 15 full time employees.

Business Value: This system will increase efficiency by eliminating manual processes. It will improve customer satisfaction and retain customers by improving how service is delivered. It will also improve better decision making through faster access to critical data. Reduce mistakes on booking equipment or personnel. Help with business growth and help prevent 5-10% loss of customers over two years.

Tangible Value:

- **Reduce labor cost in scheduling people not needed or more hours than intended**
- **Increase Revenue with faster booking and faster response time**
- **Reduce errors in double booking or outdated prices**
- **Faster proposal and turnaround**
- **Better inventory management some equipment can cost $5,000-$20,000**

Intangible Value:

- **Improved customer satisfaction and not lose 5%-10% of customers**
- **Improved decision making**

- **Support of growth of up to 15 employees**
- **Better Image on getting the job done right the first time**

Special issues (Assumptions or Risk or Constraints):

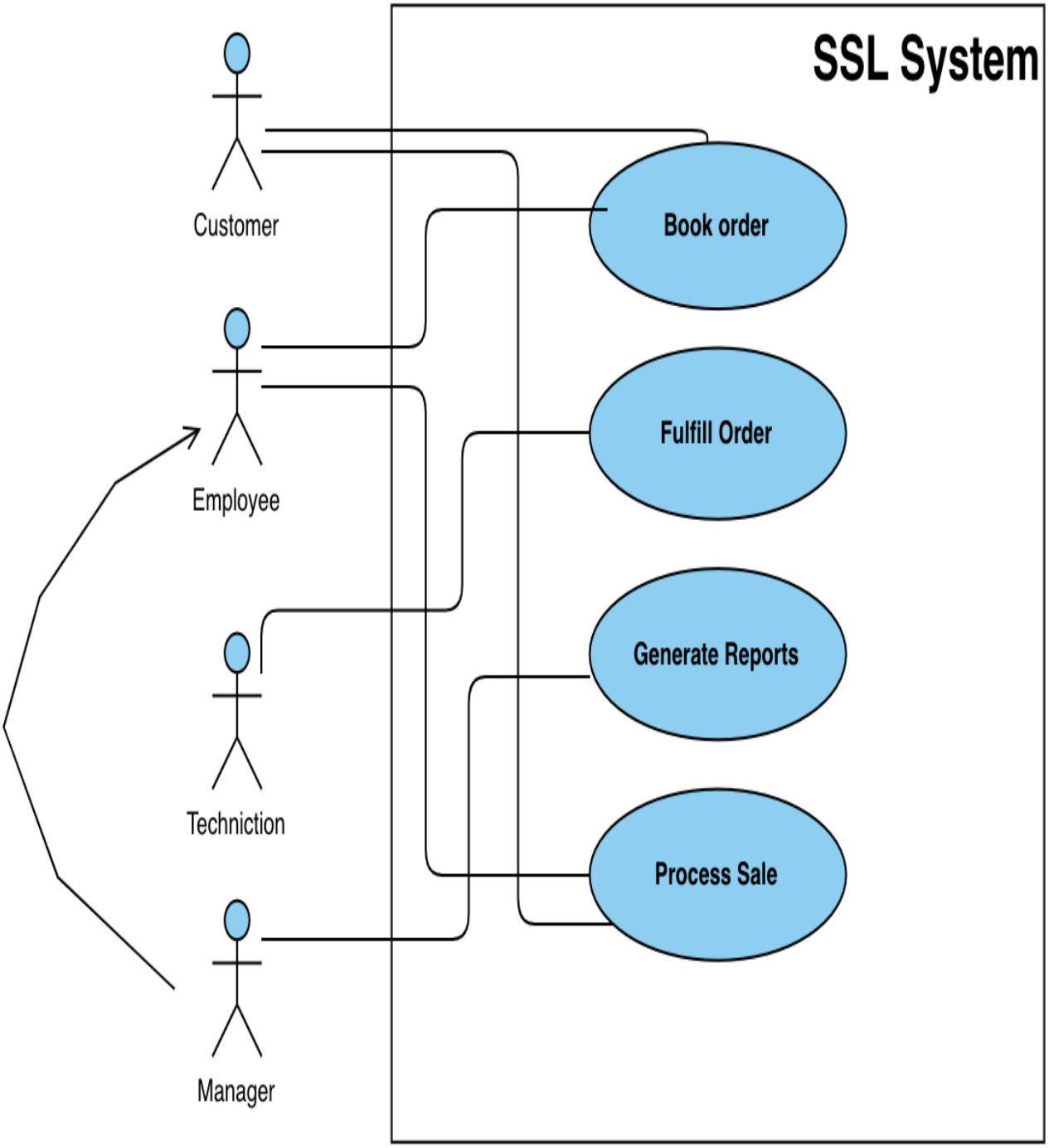
Project has a cap of $15,000. The system must be user friendly and easy to adopt by a small team. Must be able to integrate with existing tools that the company have been using. Must support remote access. Time sensitive and needs to improve customer service to prevent customer loss. Some risk is that employees might resist change. Might be errors or delays with bringing the old data into the new system. Might also have lack of training which will cause the team not to use the system properly.

Requirements Functional(FR) and Non-Functional(NF):

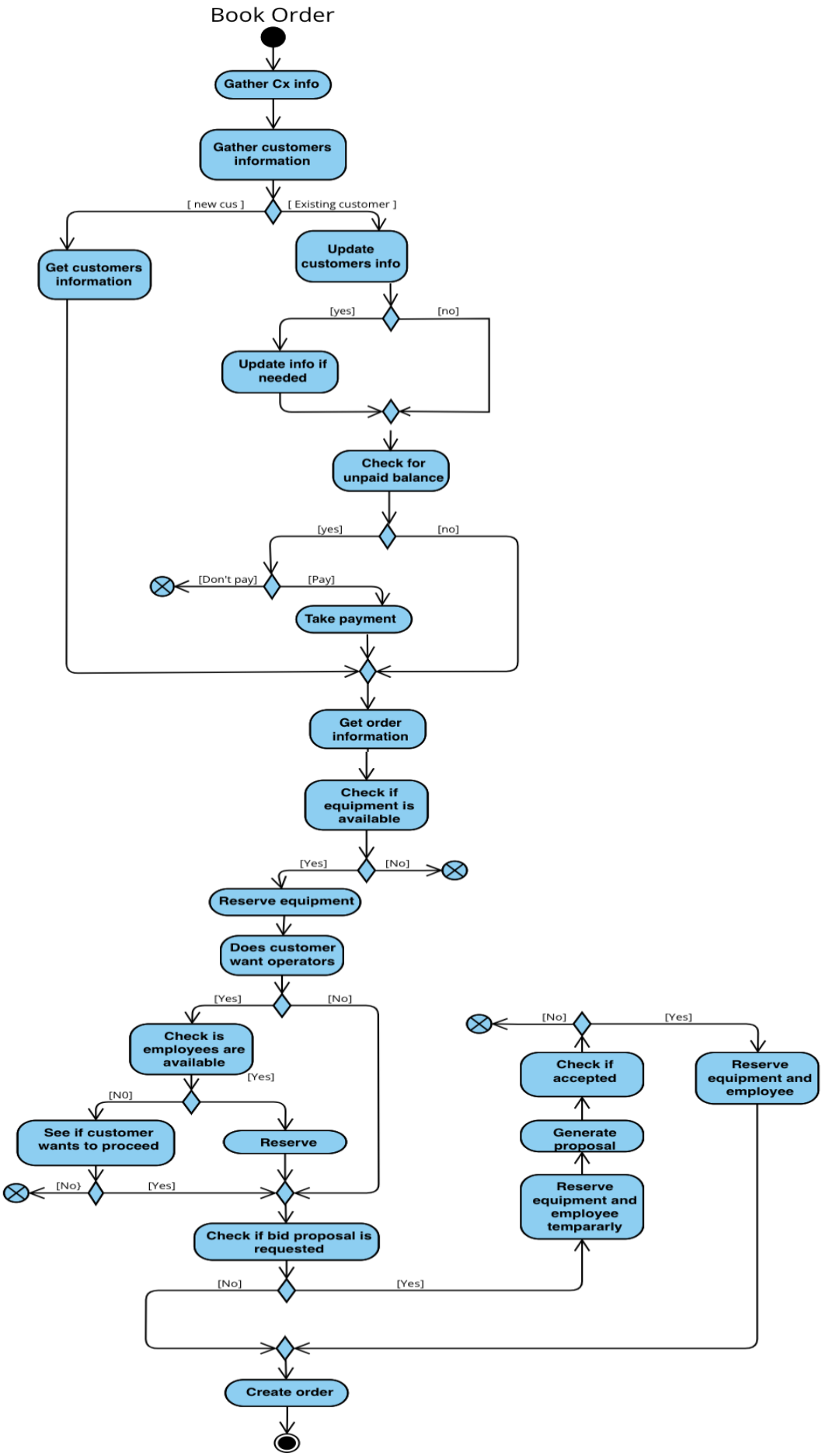
FR1: System should allow staff to create new customer profiles
FR2: System should allow updates to customer profiles
FR3: System should let team know if customer has unpaid balance
FR4: System should have customers past orders
FR5: System should allow team to search customer up with name or phone number
FR6: System should have all Available equipment for sale or rent.
FR7: System should log equipment condition or maintenance status
FR8: System should flag damage equipment
FR9: System should mark damage equipment
FR10: System should track if equipment is needed and order/rent one
FR11: System should generate customer proposals
FR12: System should check on requirements and availability
FR13: System needs to convert proposals into orders
FR14: System needs to track status of proposals status
FR15: System needs to link all proposals and orders with customers info and billing
FR16: System should hold and save all completed orders for reference.
FR17: System should generate an invoice for customer
FR18: System shall record payments in full, half or overdue.
FR19: System shall apply any promotions or discounts
FR20: System shall send reminders of payments overdue or not paid
FR21: System shall check availability of employees or contract workers
FR22: System shall Schedule employees/contract workers
FR23: System should alert when two events are booked at the same time
FR24: System shall allow customers to change the event and display open availability
FR25: System shall manage events by date, time, customer and location

NF1: System should respond to user inputs in 2 seconds
NF2: System should support up to 15 users
NF3: System should auto save data
NF4: System shall link data
NF5: System shall report any malfunctions on reports
NF6: System should report different model computers
NF7: Interface should be user friendly
NF8: System should take users less than a week to learn
NF9: System shall have tips for users as guides
NF10: System shall support of special tools to highlight important information such as overdue balance or double book events
NF11: System should be allowed to access remotely
NF12: System should have all files backed up in case of system failure
NF13: System shall have regular updates
NF14: System should store all logins and user actions
NF15: System shall require users to use secure login
NF16: System should make sure all passwords are complex
NF17: Customer information should be encrypted
NF18: System shall log user out after 10 minutes of inactivity
NF19: Different users should have roles to access more important information
NF20: System should allow users to register their computers to allow access
NF21: System should support 200+ customers
NF22: System should allow users to access for mobile phones and desktop
NF23: System should not cost more than $15,000
NF24: System should support Excel, PDF, Windows, and CSV
NF25: System should allow Admins to override others.

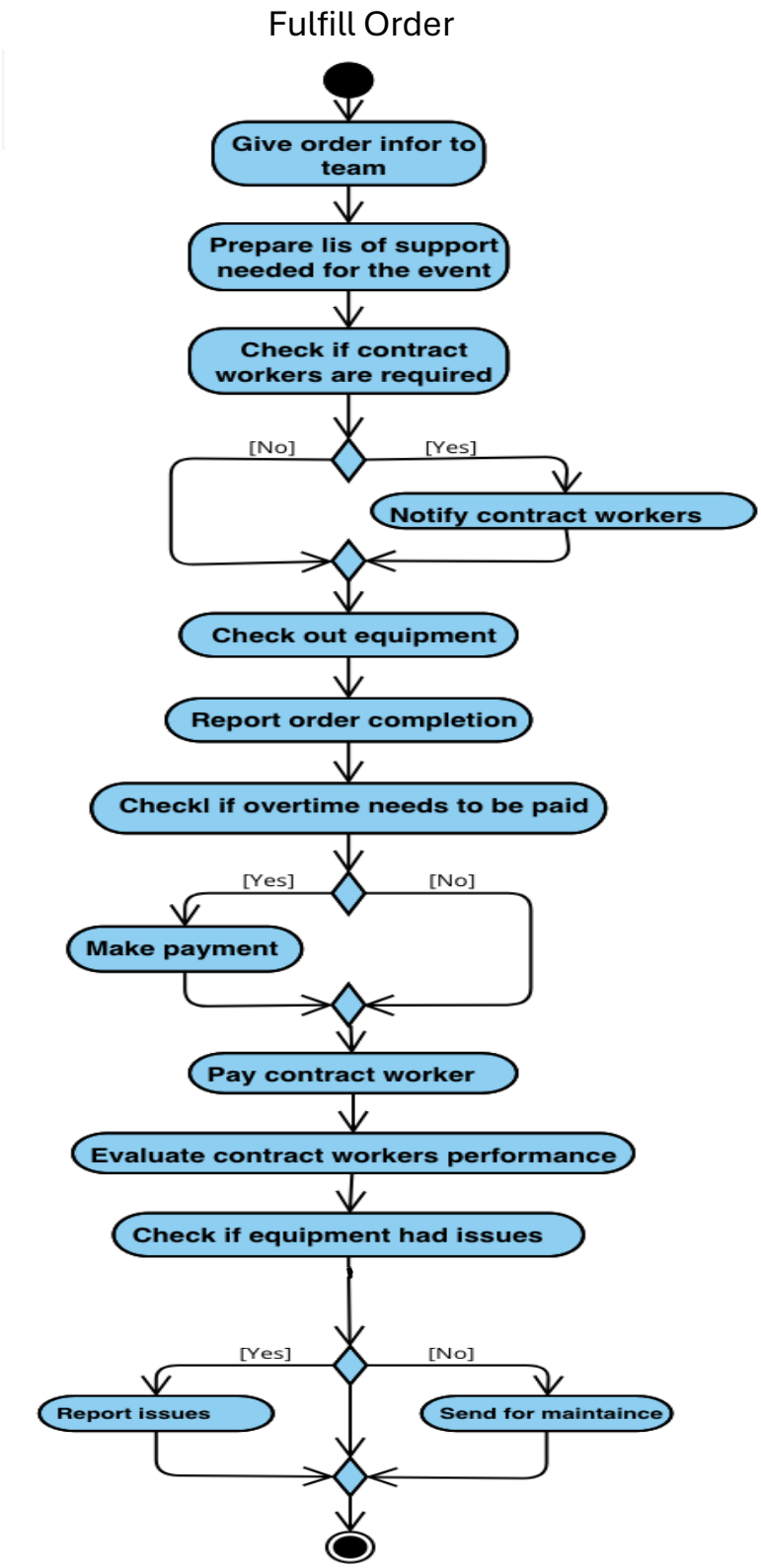
Use Case Diagram



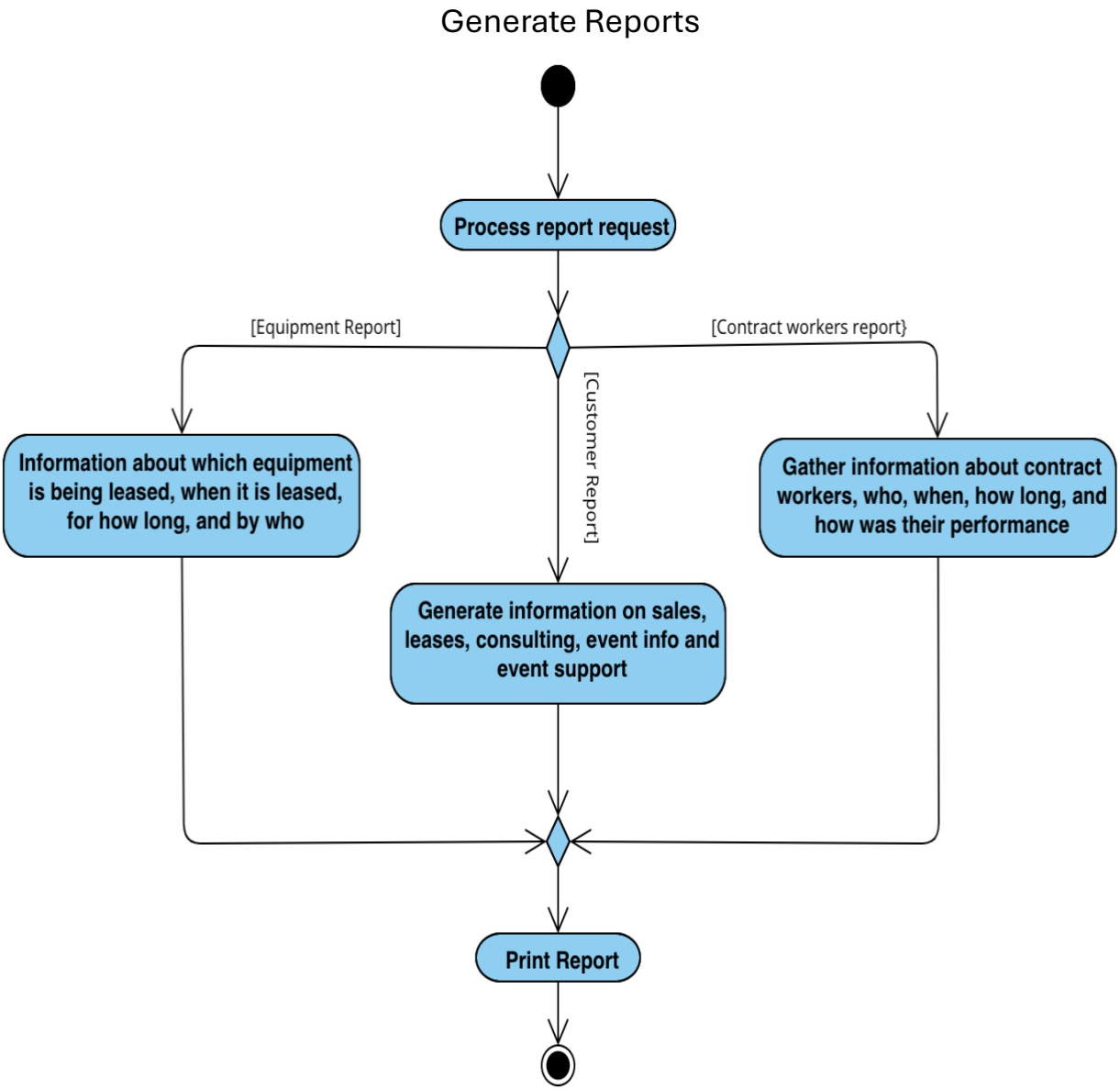
Activity Diagram 1



Activity diagram 2

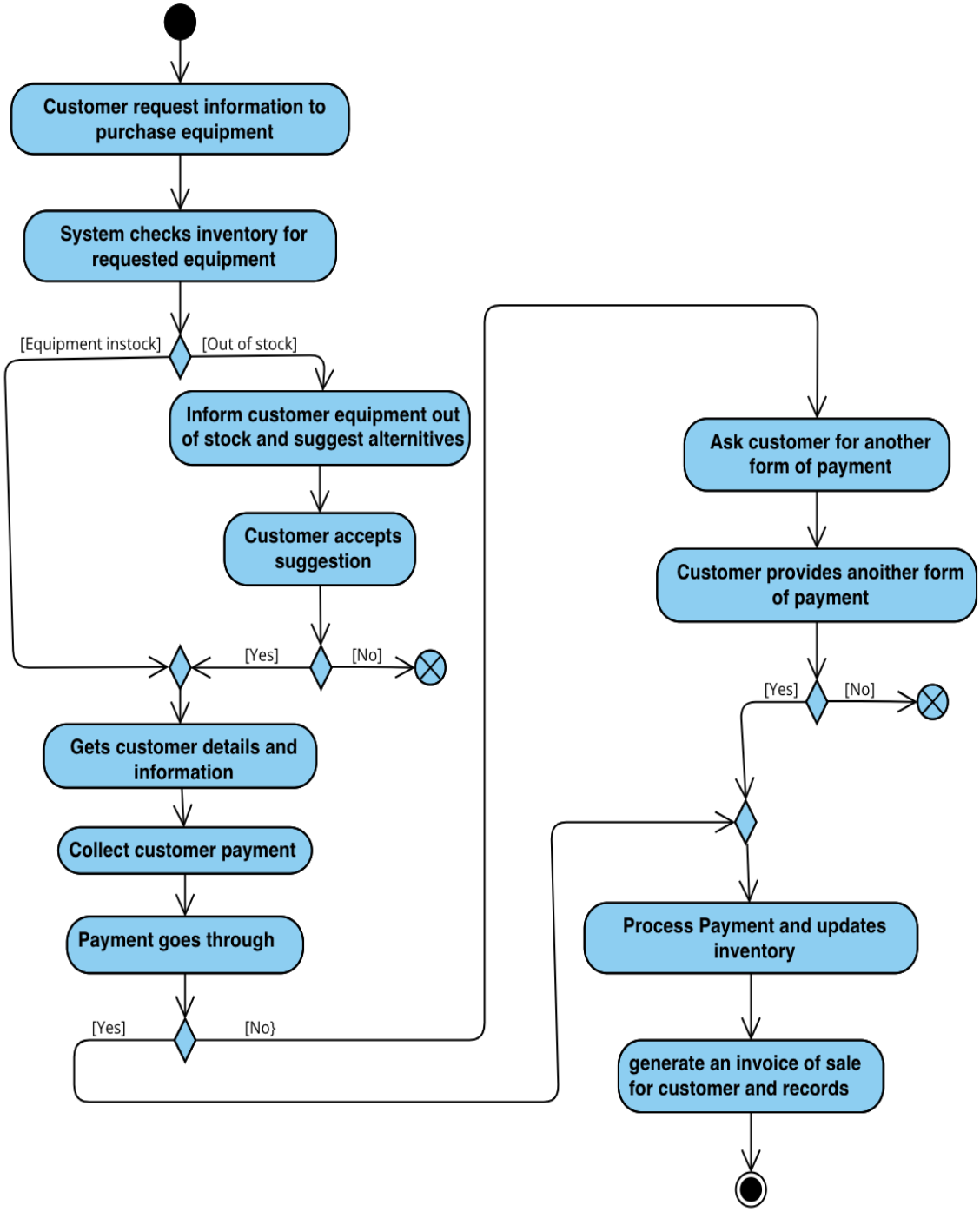


Activity diagram 3



Activity diagram 4

Process sale of equipment



Use case descriptions

Book Order: ID 1

Primary Actor: Customer

Level: High, Essential

Stakeholders and Interest: Customer wants to place order, Employee gathers information and process order

Description: A customer initiates the booking process for an event with an employee.

Relationships: Customer, Employee

Natural Flow:

1.Customer requests booking an event

2.Employee verifies customer information

3.Employee checks equipment availability for the event

4.Employee checks employee availability for the event and equipment

5. Check if proposal has been generated

6.Generates an event booking proposal

7.Sends the event booking proposal to the customer

9.Create order

Alternative flows:

2A. Customer is new

2Aa. Get customers info

2B. If existing customer check for balance

2Ba. See if customer pays

2Bb. If yes take payments

2Bc. If no end process

3A. If equipment isn't available end

4A. If available reserve and check if they need contractors

4Aa. If yes see if contractors are available

4Ab. If no end

4Ac. If yes reserve

5A. If no proposal reserve all equipment and employees and generate

5Aa. Check if accepted if no end

5Ab. If yes reserve all

Fulfilling Order: ID 2

Primary Actor: Customer

Level: High, Essential

Stakeholders and Interest: Customer wants to place order, Employee gathers information and process order

Description: Once an event booking or order is confirmed, staff prepares the required personnel and equipment for delivery or event setup.

Relationships: Employee, Technician

Main Flow:

1. Give order info to Tech
2. Prepare list of support needed for event
3. Check if contract workers are required
4. Check out equipment
5. Report order completion
6. Check if overtime needs to be paid out
7. Pay workers
8. Evaluate contract workers performance
9. Check if equipment had issues
10. Send report and end

Alternative Flow:

3A. If contract workers are required, notify them

6A. If overtime needs to be paid out pay overtime

9A. If equipment had issues send for maintenance

Generate Reports: ID 3

Primary Actor: Management

Level: High, Essential

Stakeholders and Interest: Team gathers info of the vent and sends to manager. Manager than makes report

Description: Management requests reports on business analytics, including customer history, equipment usage/history, and revenue.

Relationships: Manager, event team

Main Flow:

1. Process Report
2. Gathers equipment report, customer report and contract workers report
3. Makes final Report

Alternative Flow:

-

Process Sale of Equipment: ID 4

Primary Actor: Customer

Level: High,

Stakeholders and Interest: Customer request info on equipment to buy, Employee looks through inventory and process sale

Description: Customer calls in to place an order on an equipment they would like to purchase, employee check inventory and handles sales to sale equipment

Relationships: Customer, Employee

Main Flow:

1. Customer request to purchase equipment
2. System checks inventory
3. Get customer details
4. Collect payment
5. Process payment
6. Generate invoice

Alternative flow:

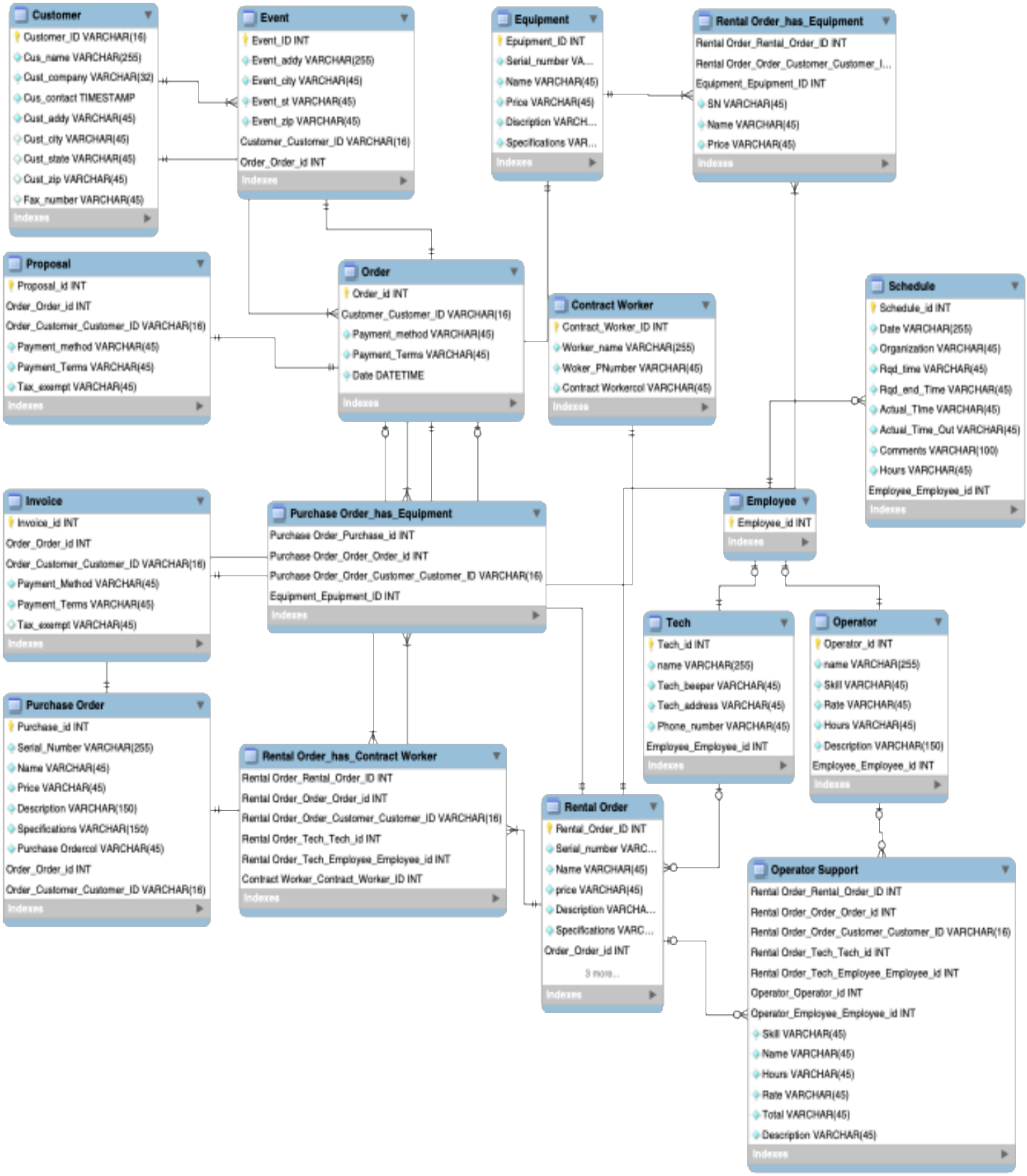
2A. Equipment out of stock suggest alternatives

3A. Customer accepts

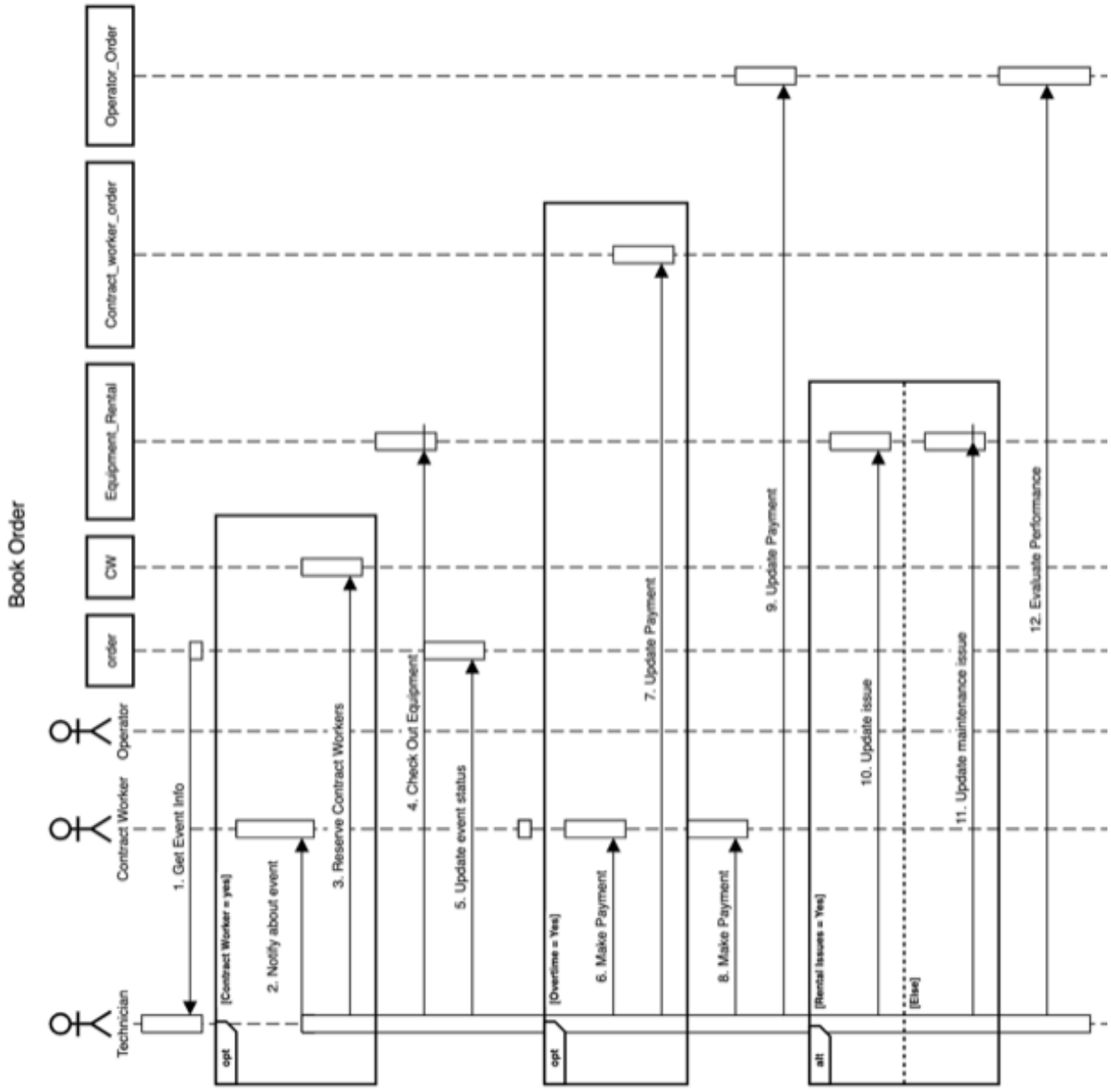
4A.Payment doesn’t go through

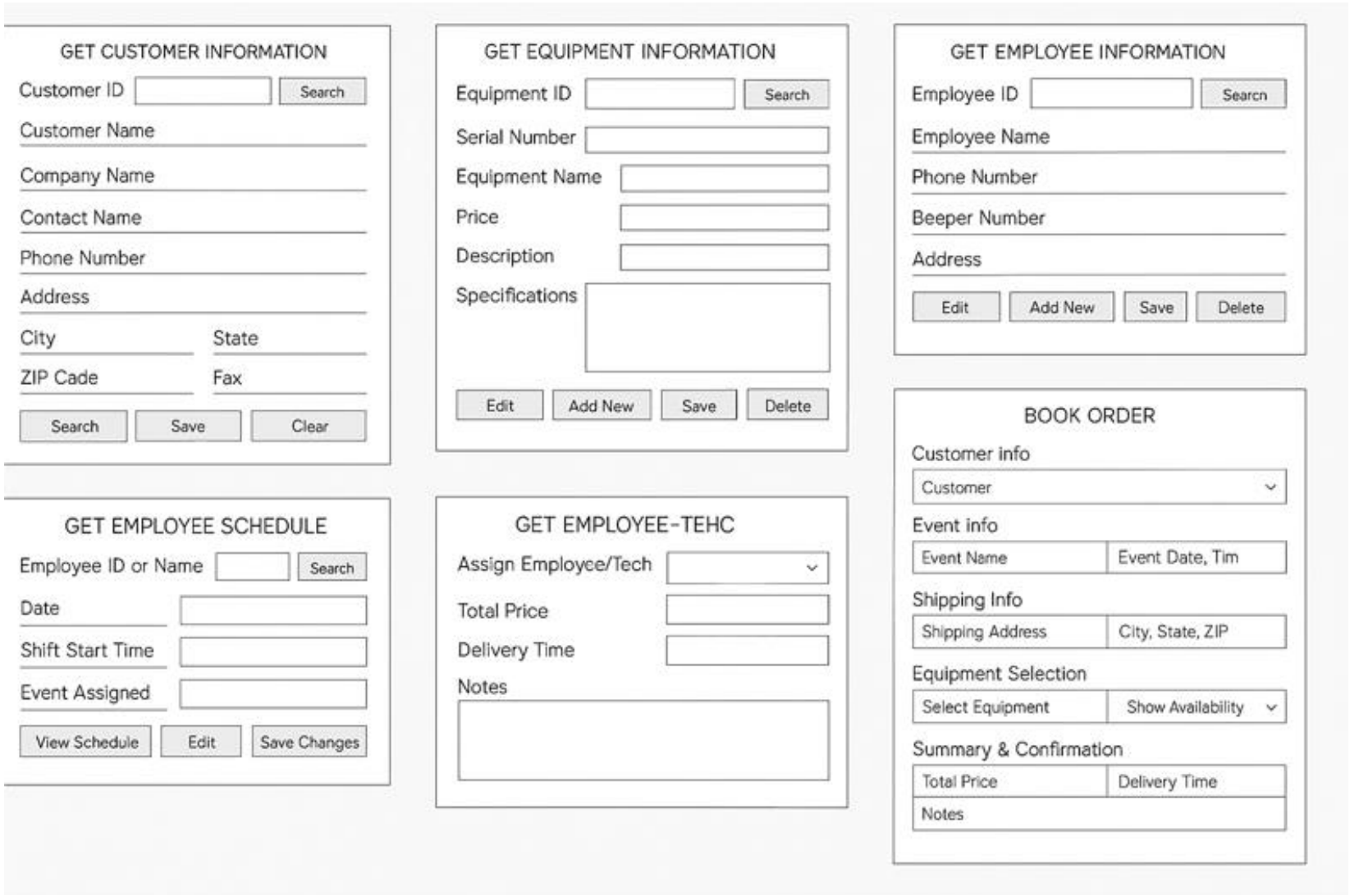
4Aa. Ask for another form of payment

ER Diagram


Customer
Customer_ID VARCHAR(16)
Cus_name VARCHAR(255)
Cust_company VARCHAR(32)
Cus_contact TIMESTAMP
Cust_addy VARCHAR(45)
Cust_city VARCHAR(45)
Cust_state VARCHAR(45)
Cust_zip VARCHAR(45)
Fax_number VARCHAR(45)
Indexes
Event
Event_ID INT
Event_addy VARCHAR(255)
Event_city VARCHAR(45)
Event_st VARCHAR(45)
Event_zip VARCHAR(45)
Customer_Customer_ID VARCHAR(16)
Order_Order_id INT
Indexes
Equipment
Equipment_ID INT
Serial_number VA...
Name VARCHAR(45)
Price VARCHAR(45)
Discription VARCH...
Specifications VAR...
Indexes
Rental Order_has_Equipment
Rental Order_Rental_Order_ID INT
Rental Order_Order_Customer_Customer_I...
Equipment_Equipment_ID INT
SN VARCHAR(45)
Name VARCHAR(45)
Price VARCHAR(45)
Indexes
Proposal
Proposal_id INT
Order_Order_id INT
Order_Customer_Customer_ID VARCHAR(16)
Payment_method VARCHAR(45)
Payment_Terms VARCHAR(45)
Tax_exempt VARCHAR(45)
Indexes
Order
Order_id INT
Customer_Customer_ID VARCHAR(16)
Payment_method VARCHAR(45)
Payment_Terms VARCHAR(45)
Date DATETIME
Indexes
Contract Worker
Contract_Worker_ID INT
Worker_name VARCHAR(255)
Woker_PNumber VARCHAR(45)
Contract Workercol VARCHAR(45)
Indexes
Schedule
Schedule_id INT
Date VARCHAR(255)
Organization VARCHAR(45)
Rqd_time VARCHAR(45)
Rqd_end_Time VARCHAR(45)
Actual_TIme VARCHAR(45)
Actual_Time_Out VARCHAR(45)
Comments VARCHAR(100)
Hours VARCHAR(45)
Employee_Employee_id INT
Indexes
Invoice
Invoice_id INT
Order_Order_id INT
Order_Customer_Customer_ID VARCHAR(16)
Payment_Method VARCHAR(45)
Payment_Terms VARCHAR(45)
Tax_exempt VARCHAR(45)
Indexes
Purchase Order_has_Equipment
Purchase Order_Purchase_id INT
Purchase Order_Order_Order_id INT
Purchase Order_Order_Customer_Customer_ID VARCHAR(16)
Equipment_Epuipment_ID INT
Indexes
Employee
Employee_id INT
Indexes
Tech
Tech_id INT
name VARCHAR(255)
Tech_beeper VARCHAR(45)
Tech_address VARCHAR(45)
Phone_number VARCHAR(45)
Employee_Employee_id INT
Indexes
Operator
Operator_id INT
name VARCHAR(255)
Skill VARCHAR(45)
Rate VARCHAR(45)
Hours VARCHAR(45)
Description VARCHAR(150)
Employee_Employee_id INT
Indexes
Purchase Order
Purchase_id INT
Serial_Number VARCHAR(255)
Name VARCHAR(45)
Price VARCHAR(45)
Description VARCHAR(150)
Specifications VARCHAR(150)
Purchase Ordercol VARCHAR(45)
Order_Order_id INT
Order_Customer_Customer_ID VARCHAR(16)
Indexes
Rental Order_has_Contract Worker
Rental Order_Rental_Order_ID INT
Rental Order_Order_Order_id INT
Rental Order_Order_Customer_Customer_ID VARCHAR(16)
Rental Order_Tech_Tech_id INT
Rental Order_Tech_Employee_Employee_id INT
Contract Worker_Contract_Worker_ID INT
Indexes
Rental Order
Rental_Order_ID INT
Serial_number VARC...
Name VARCHAR(45)
price VARCHAR(45)
Description VARCHA...
Specifications VARC...
Order_Order_id INT
3 more...
Indexes
Operator Support
Rental Order_Rental_Order_ID INT
Rental Order_Order_Order_id INT
Rental Order_Order_Customer_Customer_ID VARCHAR(16)
Rental Order_Tech_Tech_id INT
Rental Order_Tech_Employee_Employee_id INT
Operator_Operator_id INT
Operator_Employee_Employee_id INT
Skill VARCHAR(45)
Name VARCHAR(45)
Hours VARCHAR(45)
Rate VARCHAR(45)
Total VARCHAR(45)
Description VARCHAR(45)
Indexes

Sequence Diagram




GET CUSTOMER INFORMATION
Customer ID
Search
Customer Name
Company Name
Contact Name
Phone Number
Address
City
State
ZIP Cade
Fax
Search
Save
Clear
GET EQUIPMENT INFORMATION
Equipment ID
Search
Serial Number
Equipment Name
Price
Description
Specifications
Edit
Add New
Save
Delete
GET EMPLOYEE INFORMATION
Employee ID
Searcn
Employee Name
Phone Number
Beeper Number
Address
Edit
Add New
Save
Delete
GET EMPLOYEE SCHEDULE
Employee ID or Name
Search
Date
Shift Start Time
Event Assigned
View Schedule
Edit
Save Changes
GET EMPLOYEE-TEHC
Assign Employee/Tech
Total Price
Delivery Time
Notes
BOOK ORDER
Customer info
Customer
Event info
Event Name
Event Date, Tim
Shipping Info
Shipping Address
City, State, ZIP
Equipment Selection
Select Equipment
Show Availability
Summary & Confirmation
Total Price
Delivery Time
Notes